

Dual-Rocket SoC

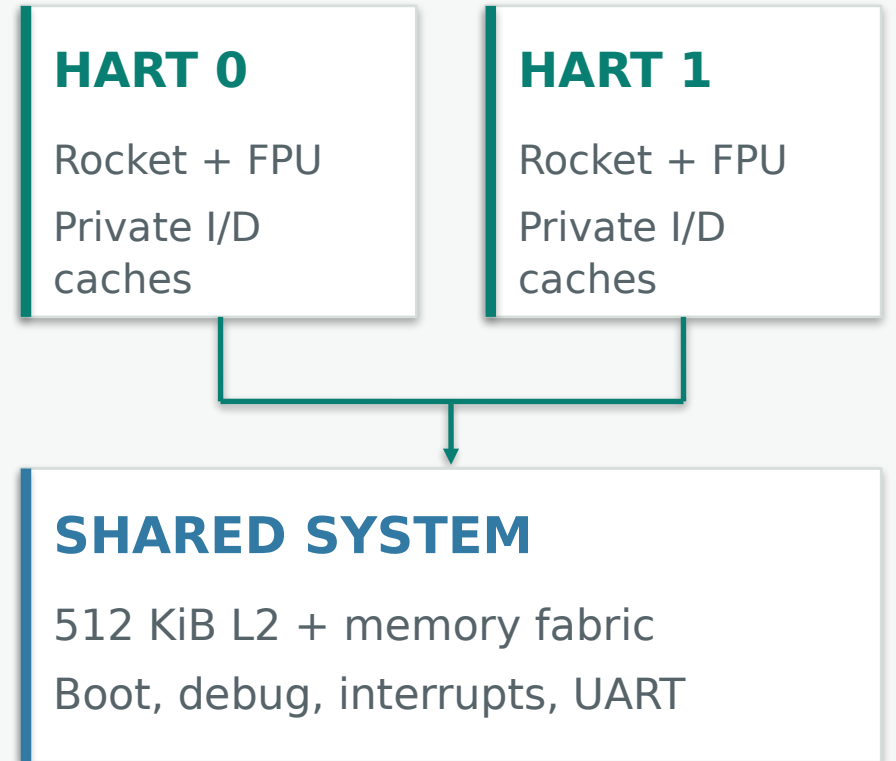
Configuration, architecture and verification

Two 64-bit RISC-V cores. One shared-memory SoC. Built from pinned Chipyard sources and evaluated with open-source tools.

Prepared for sainadh812 | 08 September 2026

Scope: generated RTL and exploratory mapping. No fabricated chip or physical signoff.

Evidence: CONFIGURATION.md | SOC.md



What does this SoC do?

A SoC combines processors, memory infrastructure and system services.

COMPUTE

Two instruction streams
Integer and floating-point work

SHARE DATA

Caches and coherent interconnect
Memory and atomic operations

CONTROL

Boot, interrupts, UART
External memory and debug

Demonstrated in simulation

- RISC-V ISA programs and scalar benchmarks
- Both-hart output, shared data and atomic updates

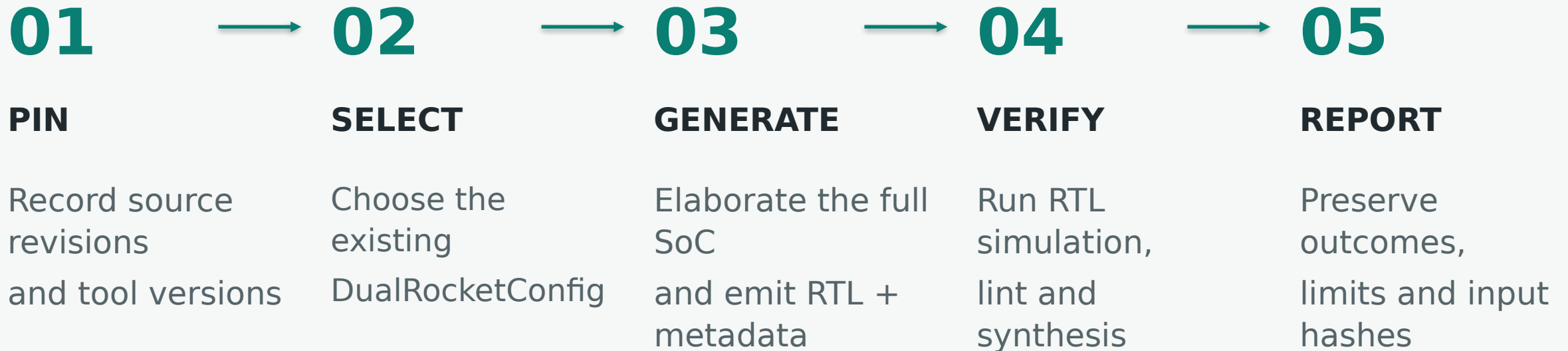
Not yet demonstrated

Custom firmware and multicore applications;
OS bring-up needs its own software and tests.

This is a development and verification platform, not a complete board-ready computer.

How the project was built

The work changed the selected configuration and tooling, not the generated CPU RTL.



Configuration and SoC documented before fresh lint/synthesis; simulation predates those reruns.

Recorded evidence includes unsuccessful attempts and warnings, not only the passing summaries.

The exact two-core configuration

Chipyard already provides this class; no hand-duplication of a Rocket core was needed.

```
class DualRocketConfig extends Config(  
  new freechips.rocketchip.rocket.WithNHugeCores(2) ++  
  new chipyard.config.AbstractConfig)
```

SINGLE-CORE BASELINE

RocketConfig uses WithNHugeCores(1).

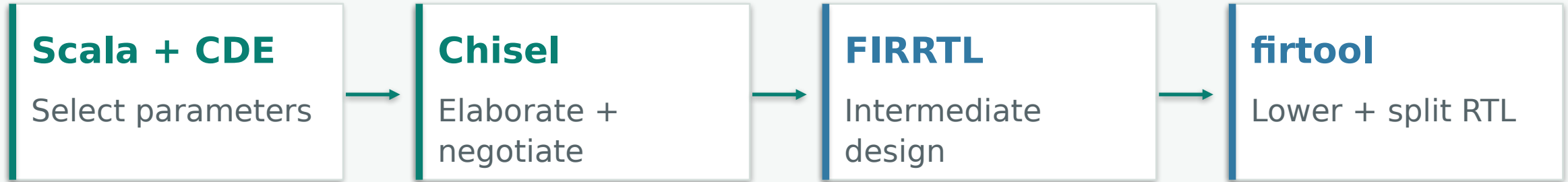
SELECTED DESIGN

DualRocketConfig uses WithNHugeCores(2).

The second hart shares the system fabric and L2; this is not two independent SoCs.

From Scala to generated hardware

The configuration is a recipe. Chipyard elaborates that recipe into RTL.



Generated outputs

- Split Verilog / SystemVerilog
- DTS, memory map, L2 metadata
- Memory interfaces and test selections

476

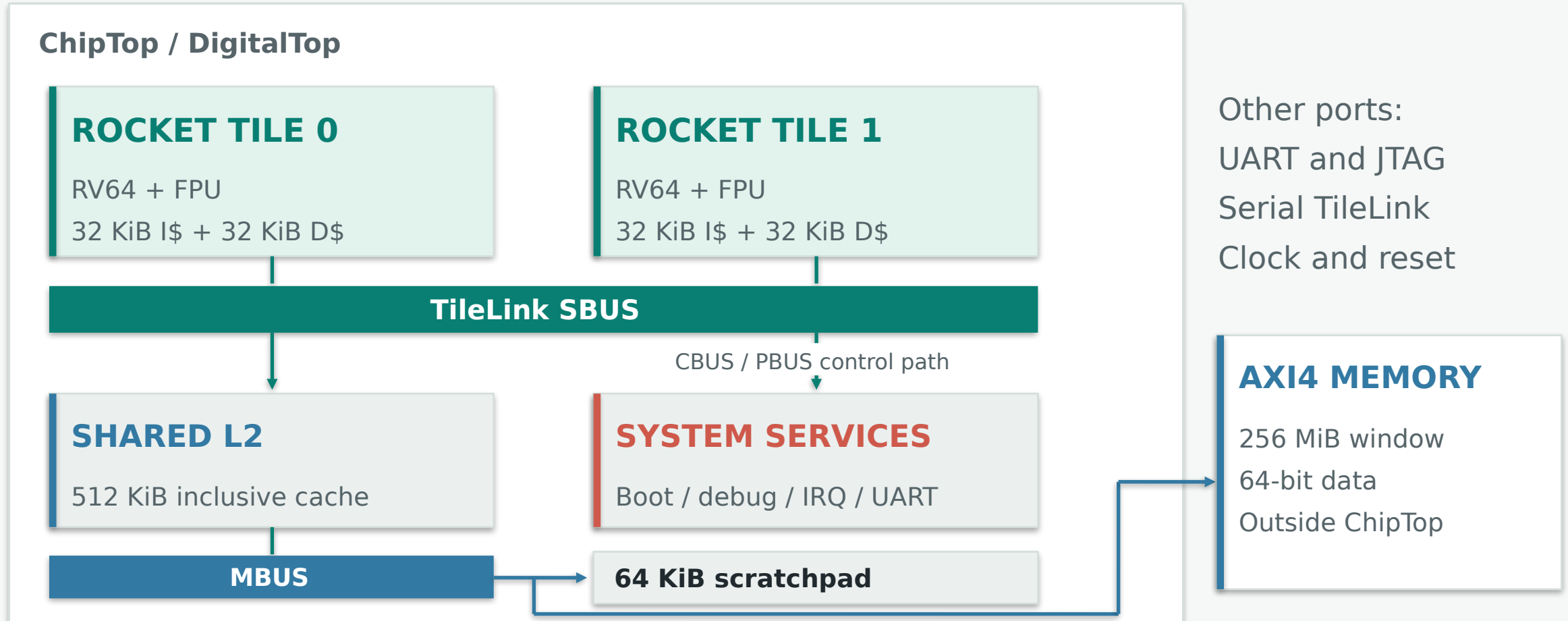
generated RTL source files

Snapshot includes simulation resources.
Only the selected hierarchy is analyzed.

RTL generation and simulator compilation are separate make targets.

Inside the full ChipTop SoC

Simplified functional grouping; individual bus adapters and control branches are omitted.



Evidence: SOC.md | artifacts/chipyard.harness.TestHarness.DualRocketConfig.l2.json

What each Rocket tile contains

Hart 0 and hart 1 use the same core preset and private L1 geometry.

RESOURCE	PER-HART CONFIGURATION
Execution	RV64, in-order, single-instruction decode / retirement
Arithmetic	I/M/A/C + Zba/Zbb/Zbs; floating point F/D/Zfh
Privilege and MMU	Machine / supervisor / user modes; Sv39 virtual memory
Instruction cache	32 KiB, 8 ways, 64 sets, 64-byte lines
Data cache	32 KiB, 8 ways; blocking cache (nMSHRs = 0)
Protection and TLBs	8 PMP entries; 32-entry instruction and data TLBs

Not selected: vector V, Zbc carry-less multiply, or Zicboz cache-block zero.

The memory hierarchy

Architectural capacities are different from mapped SRAM bits and library cell area.

PRIVATE L1

128 KiB total

32 KiB I\$ + 32 KiB D\$ per hart

SHARED L2

512 KiB

Inclusive, 8-way, one coherence bank

SCRATCHPAD

64 KiB

On-chip at 0x08000000; DTS disabled

MAIN MEMORY

256 MiB window

External AXI4 region at 0x80000000

External main memory is not on-chip DRAM. ChipTop has no DDR controller or DDR PHY.

Boot, peripherals and external interfaces

The base SoC includes essential control and debug, not every optional Chipyard device.

BOOT + INTERRUPTS

Boot ROM / boot selection
CLINT timer + software IRQ
PLIC: UART device interrupt

EXTERNAL INTERFACES

UART RX/TX and JTAG
AXI4 external memory
Serial TileLink; clocks/reset

NOT IN THIS CONFIG

FFT / RoCC / vector unit
GPIO, I2C, SPI flash
Ethernet or DDR controller

Do we have FFT?

No dedicated FFT hardware. A Rocket CPU can execute FFT software;

an FFT accelerator needs a new integration and its own verification.

Optional generator code in Chipyard does not mean that block exists in this generated SoC.

Key software-visible address regions

Addresses come from the emitted memory map and device tree.

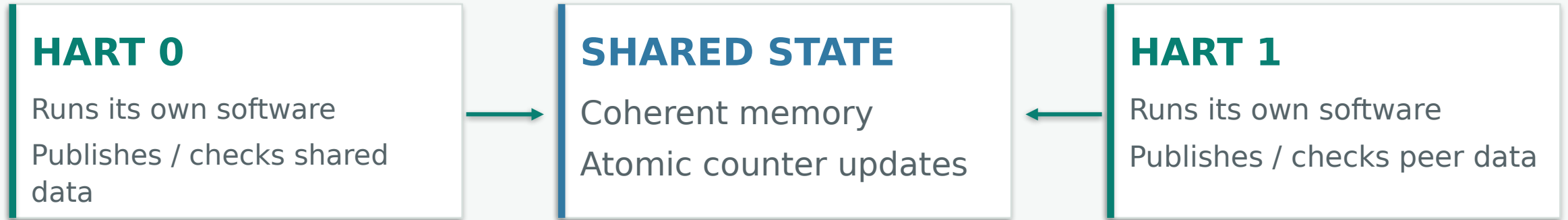
BASE ADDRESS	WINDOW	RESOURCE / PURPOSE
0x00000000	4 KiB	Debug module
0x00010000	64 KiB	Boot ROM address window
0x02000000	64 KiB	CLINT: timer / software interrupts
0x02010000	4 KiB	Shared L2 control registers
0x08000000	64 KiB	On-chip scratchpad
0x0c000000	64 MiB	PLIC register aperture
0x10020000	4 KiB	UART registers
0x80000000	256 MiB	External main-memory window

Register-window size is not storage capacity. Full address table is in SOC.md.

Evidence: SOC.md | artifacts/chipyard.harness.TestHarness.DualRocketConfig.memmap.json

How two cores cooperate

Separate instruction streams can coordinate through shared memory and atomic operations.



2 harts

required by explicit smoke

256

total atomic increments

PASS

peer-data visibility checked

Software must use both harts; two cores do not automatically make every program parallel.

Hardware boundary vs simulation harness

Full-SoC synthesis means ChipTop, not the whole simulated environment.

TestDriver / TestHarness: simulation environment

ChipTop / DigitalTop

Both Rocket tiles, L1s and L2
Scratchpad, buses and peripherals
Clock/reset logic and I/O cells

EXTERNAL TEST MODELS

DRAMSim2 / simulated memory
SimJTAG, serial host, UART output
Test clocks and fast ELF loading

Synthesis boundary = ChipTop. +loadmem is a simulation shortcut, not fabricated hardware.

Three checks answer different questions

Common source evidence; separate methods, outputs and acceptance boundaries.

SIMULATION

Does this test execute?

Verilator + DRAMSim2
RISC-V programs and multicore tests
Reference checks with Spike

LINT

What does static analysis flag?

Verilator + slang
Full-ChipTop diagnostics
Warning baseline and source checks

SYNTHESIS

Can the design be mapped?

Yosys / slang + ABC
SRAM adapters and libraries
OpenROAD netlist linking

All three are useful. None alone establishes functional completeness or manufacturing readiness.

Evidence: [reports/simulation/README.md](#) | [reports/lint/README.md](#)

Simulation: configured tests passed

Recorded full-SoC RTL campaign, completed 06 September 2026. Not rerun for this deck.

335 / 335

configured ISA tests

12 / 12

configured benchmarks

Broader inventory: 422 unique programs / variants



391 reported PASS

25 FAIL

6 wall timeouts

The expanded inventory did not pass completely. Do not report 422/422 passing.

Evidence: [reports/simulation/README.md](#) | [reports/simulation/summary.json](#)

Interpreting failures, timeouts and passes

Test intent and the correctness of the test itself matter.

9 ISA NON-PASSES

Additional unsupported-feature or misaligned-access expectation probes

22 LEGACY NON-PASSES

Matrix geometry / test-oracle defects;
32 x 32 assumptions vs 16 x 16 data

- Some reported legacy passes also have bounds or result-checking caveats.
- Three corrected multicore supplements passed both RTL and Spike.
- Spike agreed on 391 PASS and 31 non-PASS classifications.

Agreement with Spike is supporting evidence, not formal equivalence or exhaustive verification.

Evidence: [reports/simulation/README.md](#) | [reports/simulation/reference-comparison/summary.json](#)

Lint: zero errors, warnings retained

Both tools analyzed full ChipTop using the original behavioral memories.

TOOL	ERRORS	WARNINGS	NOTES
Verilator 5.051	0	2,484	0
slang 11.0.448	0	21	21

Dominant baseline

2,047 unused-signal warnings
424 empty pin connections

Review items still visible

4 reset-related SYNCASYNCCNET warnings
1 clock-gate latch; 21 shift expressions

Zero errors is not warning-free signoff. Dedicated CDC/RDC review remains necessary.

Evidence: reports/lint/README.md | reports/lint/evidence/summary.json

How full-SoC synthesis was performed

Yosys performs synthesis and mapping; OpenROAD checks the mapped netlist can link.

01 Validate RTL + SRAM inventory

Check dimensions, masks, instances and source hashes.

02 Check SRAM adapters

Compare generated memory behavior over seven interfaces.

03 Synthesize and technology-map

Yosys + slang + ABC; Nangate45 / fakeram45 libraries.

04 Link and audit the result

OpenROAD reads all mapped instances; recheck inputs.

Technology mapping completed. No timing constraint, floorplan, placement or routing was run.

SRAM adaptation was a separate task

Logical memories must be matched to available macro width, depth and write-mask behavior.

7

interface shapes

16

logical memory instances

194

512 x 64 SRAM macros

Two-core-specific change

L2 directory: 1024 x 144 bits

Mask granularity: 18 bits

Adapter comparison result

7 interfaces x 3 seeds

212,364 compared cycles: PASS

Finite initialized-state simulation passed. Formal and X-state equivalence were not established.

Evidence: [reports/synthesis/README.md](#) | [reports/synthesis/inventory.json](#)

Synthesis results: mapped and linked

Exploratory Nangate45 / fakeram45 library estimate. Not die area or manufacturing signoff.

383,763

total mapped instances

383,569

standard cells

194

SRAM macros

4.004733 mm²

total Liberty cell-area sum



SRAM: 3.356479 mm² (83.8%)

Standard cells: 0.648254 mm²

55,694 flip-flops + 1 latch are included in standard cells. Unmapped cells / memories / processes: 0.

Evidence: reports/synthesis/README.md | reports/synthesis/mapped-statistics.json

What is not proven yet

Successful elaboration, tests and mapping are milestones, not chip signoff.

AREA	REMAINING WORK / BOUNDARY
Clock and timing	500 MHz is nominal intent, not achieved CPU frequency.
Physical design	No placement, routing, power analysis or DRC/LVS.
Functional closure	No exhaustive coherence, formal proof or coverage percentage.
Software / debug	Linux boot and full JTAG/OpenOCD workflows not tested.
Serial and reset	Heavy-traffic serial-PHY warning; reset-domain review needed.
Mapped behavior	No full-SoC SRAM-mapped gate-level regression.

Report the design as a verified experiment with explicit limits, not a production-ready ASIC.

Evidence: SOC.md | reports/simulation/README.md

How to try the next SoC configuration

Proposed work only. None of these changes is included in the current generated SoC.

OPTION A: CORE / CACHE

Change core count or cache geometry.
Recheck coherence clients and SRAM shapes.

OPTION B: FFT ACCELERATOR

Integrate the optional generator.
Define interface and numerical tests.

- Use a new named Config and preserve this baseline.
- Regenerate RTL, device tree, memory map and SRAM inventory.
- Add targeted software tests, then rerun lint and full-SoC synthesis.

Choose the workload first; compare new designs with the same tools and explicit acceptance criteria.

Where the reproducible process lives

The results repository is not a complete toolchain or a vendored Chipyard checkout.

```
bash scripts/dual-rocket/generate.sh
bash scripts/dual-rocket/lint/run.sh
bash scripts/dual-rocket/synthesis/run.sh
```

Architecture + process

docs/dual-rocket/CONFIGURATION.md
docs/dual-rocket/SOC.md

Evidence + portable scripts

docs/dual-rocket/reports/
scripts/dual-rocket/README.md

Important: the top-level rtl/ folder is historical single-core RTL, not this dual-core build.

Evidence: CONFIGURATION.md | scripts/dual-rocket/README.md

Pinned tools and source provenance

Versions describe the recorded experiment; this is not a recommendation to use arbitrary latest builds.

LAYER	RECORDED INPUT
Chipyard / Rocket Chip	e602d917dcc4 / 55bcad0f5943 (full SHA in notes)
Generator stack	Chisel 6.7.0; Scala 2.13.16; firtool 1.75.0
Java	OpenJDK 20.0.2-internal
RTL simulation	Verilator 5.022; DRAMSim2; matched Spike
Lint	Verilator 5.051 devel; slang 11.0.448
Mapping / memory checks	Yosys 0.68+195; ABC; Icarus Verilog 14.0 devel
Netlist link / libraries	OpenROAD v2.0-17598; pinned ORFS Nangate45 / fakeram45

476 RTL source hashes preserved; all 500 captured synthesis input hashes rechecked unchanged.

Evidence: CONFIGURATION.md | reports/simulation/README.md

Quick glossary and evidence guide

All slides have speaker notes with project-relative source references.

TERM	MEANING IN THIS PROJECT
SoC / hart	Integrated system / one hardware execution thread
RTL / Config	Hardware description / generator parameter composition
L1 / L2	Private per-core caches / shared inclusive cache
TileLink / AXI4	On-chip fabric / external-memory interface
CLINT / PLIC	Timer + software interrupts / device interrupts

Start reading: docs/dual-rocket/README.md

Bottom line: two Rocket cores, one complete SoC, reproducible evidence and clearly stated limits.

Evidence: SOC.md | CONFIGURATION.md